

Abdullah Chabaytah

514-705-5932 | a.chabaytah@mail.utoronto.ca | [linkedin.com/in/achabaytah](https://www.linkedin.com/in/achabaytah) | github.com/maniic | maniic.github.io

EDUCATION

University of Toronto

Toronto, ON

HBSc Computer Science, Math & Statistics

Expected Grad: 05/2028

- **Relevant Coursework:** *Data Structures & Algorithms, Theoretical Computation, Software Design, Systems Programming, Multivariable Calculus, Calculus with Proofs, Probability, Statistics & Data Analysis*

EXPERIENCE

Sports Up

05/2025 – 09/2025

Software Developer Intern

Toronto, ON

- Engineered a cross-platform mobile application using **React Native & Firebase**, achieving **99% uptime** across **100+ users** under sustained loads of **50 requests/sec** via real-time data synchronization & optimized query batching
- **Reduced crash rates by 35% & increased weekly active users by 20%** by refactoring backend architecture and rebuilding UI/UX components to eliminate state management bottlenecks
- Implemented end-to-end testing frameworks using automated test suites to enforce deployment quality gates, cutting regression failures across 3 major feature releases

Outlier AI

01/2025 – Present

Data Labeling Contractor

Remote

- Annotated & validated **NLP** and **computer vision** training datasets, applying domain-specific labeling schemas to improve model precision across production ML pipelines

PROJECTS

Predictive Maintenance | *Python, PyTorch, MLflow, Scikit-learn, Pandas*

- Trained 7 deep learning architectures including **Bidirectional LSTM, Temporal CNN, and Transformer** using **PyTorch Lightning** to predict jet engine failure on **NASA** benchmark data, achieving **11.71 RMSE**
- Built end-to-end data pipelines processing 21 sensor channels across multiple fault-mode datasets, implementing normalization, sequence windowing, and automated data quality validation
- Instrumented **MLflow** to log hyperparameters, metrics, and model artifacts across **100+** training runs, enabling reproducible experiment tracking and model comparison

Sentiment Forecasting | *Python, PyTorch, Hugging Face, Scikit-learn, Pandas*

- Built a quantitative forecasting pipeline achieving a **2.34 Sharpe ratio** and **-2.5% max drawdown** over **180-day** backtests by fusing **FinBERT** outputs with **9** topological and **6** momentum features
- Trained and evaluated **Logistic Regression, XGBoost, and Ensemble** classifiers using time-series aware cross-validation to eliminate look-ahead bias, selecting the optimal model via **Sharpe ratio** on held-out data
- Implemented **Topological Data Analysis** using 4-dimensional delay embeddings on 60-day rolling windows to extract market regime signals absent from traditional indicators

MoodVerse | *Java, Stanford CoreNLP*

- Built a desktop diary system with **Stanford CoreNLP**-powered sentiment analysis, classifying emotional tone across **150+** journal entries with **95%** consistency & mapping outputs to a curated media recommendation engine
- Designed a sentiment-to-media mapping layer integrating the Spotify & TMDb APIs to surface emotion-aligned content recommendations, increasing prototype engagement with suggested media by **35%**

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C/C++, Java, Golang, R, Kotlin, Assembly, Rust, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, TensorFlow, PyTorch, FastAPI, Flask, Django, Pandas, NumPy, Matplotlib

Systems & Tools: Docker, PostgreSQL, Redis, Git, Unix/Linux, CI/CD, VS Code, PyCharm, IntelliJ